

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0939

North Bonneville 100 kV
December 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0939. The Interconnection Customer proposes a 46.9 MW generating facility interconnecting at the North Bonneville 100 kV substation in the Illinois area. The request was received on February 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 287 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,663,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Oxbow – N Bonneville 100.0 kV L, the N Bonneville – Lake Charles Plant 100.0 kV loads to 280.5% of its 114 MVA rating. Project DFAX: 100.0%.

Lake Charles Plant – Clintmoore 100.0 kV L: 137.7% loading (179 MVA) following the outage of the Oxbow – N Bonneville 100.0 kV; DFAX = 100.0%.

For the loss of St. Lawrence – Jame De Young 230.0 kV, the Adirondack – James De Young 230.0 kV loads to 101.4% of its 1,172 MVA rating. Project DFAX: 7.4%.

Myrtle Pt – George Neal S 230.0 kV: 101.2% loading (974 MVA) following the outage of the Galion Generating Sta – Trego 500.0 kV; DFAX = 6.5%.

For the loss of Trego – Galion Generating Sta 500.0 kV line, the Zona Libre 6007 – George Neal S Substation 230.0 kV loads to 101.0% of its 1,001 MVA rating. Project DFAX: 6.5%.

The Zona Libre 6007 – Kampsville 230.0 kV is overloaded to 100.8% of its 1,028 MVA emergency rating for the loss of the Galion Generating Sta – Trego 500.0 kV. The project contributes 6.5% DFAX (3.0 MW) to this overload.

The Baraga Vlg – Gray'S Flat 230.0 kV is overloaded to 100.8% of its 1,355 MVA emergency rating for the loss of the Madeline – E Methuen 230.0 kV. The project contributes 8.9% DFAX (4.2 MW) to this overload.

Moreno Tap – Kampsville 230.0 kV: 100.8% loading (1,044 MVA) following the outage of the Trego – Galion Generating Station 500.0 kV; DFAX = 6.5%.

The Gray'S Flat – E Methun 230.0 kV is overloaded to 100.7% of its 1,373 MVA emergency rating for the loss of the E Methuen – Madeline 230.0 kV. The project contributes 8.9% DFAX (4.2 MW) to this overload.

The Eastown Rd – Moreno Tap 230.0 kV Line is overloaded to 100.6% of its 1,070 MVA emergency rating for the loss of the Trego – Galion Generating Sta 500.0 kV Line. The project contributes 6.5% DFAX (3.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Bonneville – Lake Charles Plant 100.0 kV	Rebuild line with higher-capacity conductor	\$2,872,000	15% (\$419,000)
Lake Charles Plant – Clintmoore 100.0 kV L	Rebuild line with higher-capacity conductor	\$6,725,000	15% (\$981,000)
Adirondack – James De Young 230.0 kV	Reconductor line	\$12,938,000	0% (\$31,000)
Myrtle Pt – George Neal S 230.0 kV	Reconductor line	\$6,294,000	0% (\$16,000)
Zona Libre 6007 – George Neal S Substation 230.0 kV	Reconductor line	\$15,066,000	0% (\$39,000)
Zona Libre 6007 – Kampsville 230.0 kV	Reconductor line	\$14,863,000	0% (\$39,000)
Baraga Vlg – Gray'S Flat 230.0 kV	Reconductor line	\$13,886,000	0% (\$33,000)
Moreno Tap – Kampsville 230.0 kV	Reconductor line	\$7,078,000	0% (\$19,000)
Gray'S Flat – E Methun 230.0 kV	Reconductor line	\$12,902,000	0% (\$30,000)
Eastown Rd – Moreno Tap 230.0 kV Line	Reconductor line	\$21,384,000	0% (\$56,000)
	Total allocated to project		\$1,663,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.